Lab 5: Web Crawler for Product Information

Information Retrieval (417524B) - SPPU BE AI&DS

Build the web crawler to pull product information and links from an e-commerce website.

CODE

```
import warnings
warnings.filterwarnings("ignore")
import requests
from bs4 import BeautifulSoup
from urllib.parse import urljoin

base = "http://books.toscrape.com/catalogue/"
count = 0
for page in range(1, 4):
    url = f"{base}page-{page}.html"
    html = requests.get(url, timeout=20).text
    soup = BeautifulSoup(html, "html.parser")
    for item in soup.select("article.product_pod"):
        title = item.h3.a["title"]
        price = item.select_one("p.price_color").text
        link = urljoin(url, item.h3.a["href"])
        count += 1
        print(f"{count}. {title} | {price} | {link}")
        if count >= 10:
            break
    if count >= 10:
        break

print(f"\nTotal products crawled: {count}")

# base = "https://books.toscrape.com/catalogue/"
```

OUTPUT

```
1. A Light in the Attic | Â£51.77 | http://books.toscrape.com/catalogue/a-light-in-the-attic_1000/index.html
2. Tipping the Velvet | Â£53.74 | http://books.toscrape.com/catalogue/tipping-the-velvet_999/index.html
3. Soumission | Â£50.10 | http://books.toscrape.com/catalogue/soumission_998/index.html
4. Sharp Objects | Â£47.82 | http://books.toscrape.com/catalogue/sharp-objects_997/index.html
5. Sapiens: A Brief History of Humankind | Â£54.23 | http://books.toscrape.com/catalogue/sapiens-a-brief-history-of-humankind_996/index.html
6. The Requiem Red | Â£22.65 | http://books.toscrape.com/catalogue/the-requiem-red_995/index.html
7. The Dirty Little Secrets of Getting Your Dream Job | Â£33.34 | http://books.toscrape.com/catalogue/the-dirty-little-secrets-of-getting-your-dream-job_994/index.html
8. The Coming Woman: A Novel Based on the Life of the Infamous Feminist, Victoria Woodhull | Â£17.93 | http://books.toscrape.com/catalogue/the-coming-woman-a-novel-based-on-the-life-of-the-infamous-feminist-victoria-woodhull_993/index.html
9. The Boys in the Boat: Nine Americans and Their Epic Quest for Gold at the 1936 Berlin Olympics | Â£22.60 | http://books.toscrape.com/catalogue/the-boys-in-the-boat-nine-americans-and-their-epic-quest-for-gold-at-the-1936-berlin-olympics_992/index.html
10. The Black Maria | Â£52.15 | http://books.toscrape.com/catalogue/the-black-maria_991/index.html

Total products crawled: 10
```